1. The SI unit of amount of substance is

A) Mole

B) Gram

C) Kilogram

D) Liter

Answer: A

---

2. The atomic number represents

A) Number of neutrons

B) Number of protons

C) Number of electrons only

D) Mass number

Answer: B

---

3. Avogadro number is

A) 6.022×10^{23}

B) 3.022×10^{23}

C) 1.022×10^{23}

D) 9.022×10^{23}

Answer: A

---

4. pH value of neutral solution is

A) 5

B) 6

C) 7

D) 8

Answer: C

---

5. The gas used in balloons is

A) Oxygen

B) Nitrogen

C) Helium

D) Hydrogen

Answer: C

---

6. The lightest element is

A) Oxygen

B) Hydrogen

C) Helium

D) Nitrogen

Answer: B

---

7. Water is a

A) Element

B) Compound

C) Mixture

D) Solution

Answer: B

---

8. The chemical formula of sodium chloride is

A) NaCl

B) Na_2Cl

C) $NaCl_2$

D) Na_2Cl_2

Answer: A

---

9. The process of solid changing to gas is

A) Melting

B) Sublimation

C) Condensation

D) Freezing

Answer: B

10. The unit of concentration is

A) mol/L

B) kg

C) m

D) s

Answer: A

---

11. Oxidation is

A) Gain of electrons

B) Loss of electrons

C) Gain of neutrons

D) Loss of protons

Answer: B

---

12. Reduction is

A) Loss of electrons

B) Gain of electrons

C) Gain of protons

D) Loss of neutrons

Answer: B

---

13. Catalyst

A) Increases rate

B) Decreases rate

C) Stops reaction

D) Changes product

Answer: A

---

14. The atomic mass is

A) Protons + Neutrons

B) Protons only

C) Neutrons only

D) Electrons only

Answer: A

---

15. The most abundant gas in atmosphere is

A) Oxygen

B) Nitrogen

C) Carbon dioxide

D) Hydrogen

Answer: B

---

16. The SI unit of pressure is

A) Pascal

B) Newton

C) Joule

D) Watt

Answer: A

---

17. Boyle's law states

A) $P \propto V$

B) $P \propto 1/V$

C) $P \propto T$

D) $V \propto T$

Answer: B

---

18. Charles law states

A) $V \propto T$

B) $P \propto T$

C) $V \propto 1/T$

D) $P \propto 1/T$

Answer: A

---

19. The hardest natural substance is

A) Gold

B) Iron

C) Diamond

D) Silver

Answer: C

---

20. The pH of acidic solution is

A) >7

B) <7

C) =7

D) 14

Answer: B

---

21. The pH of basic solution is

A) <7

B) =7

C) >7

D) 0

Answer: C

---

22. The element with atomic number 1 is

A) Helium

B) Hydrogen

C) Lithium

D) Carbon

Answer: B

---

23. Periodic table was developed by

A) Dalton

B) Rutherford

C) Mendeleev

D) Bohr

Answer: C

---

24. Noble gases are

A) Reactive

B) Inert

C) Toxic

D) Metallic

Answer: B

---

25. The unit of molarity is

A) mol/L

B) mol/kg

C) kg/L

D) g/mol

Answer: A

---

26. Acid turns blue litmus to

A) Red

B) Green

C) Yellow

D) Blue

Answer: A

---

27. Base turns red litmus to

A) Blue

B) Red

C) Yellow

D) Green

Answer: A

---

28. HCl is

A) Base

B) Acid

C) Salt

D) Metal

Answer: B

---

29. NaOH is

A) Acid

B) Salt

C) Base

D) Gas

Answer: C

---

30. CO_2 is

A) Acidic oxide

B) Basic oxide

C) Neutral oxide

D) Metal oxide

Answer: A

---

31. The valency of oxygen is

A) 1

B) 2

C) 3

D) 4

Answer: B

---

32. The valency of hydrogen is

A) 1

B) 2

C) 3

D) 4

Answer: A

---

33. The bond formed by sharing electrons is

A) Ionic

B) Covalent

C) Metallic

D) Hydrogen

Answer: B

---

34. The bond formed by transfer of electrons is

A) Covalent

B) Ionic

C) Metallic

D) Hydrogen

Answer: B

---

35. The SI unit of energy is

A) Joule

B) Watt

C) Newton

D) Pascal

Answer: A

---

36. Electrolysis involves

A) Chemical to electrical

B) Electrical to chemical

C) Heat to electrical

D) Light to heat

Answer: B

---

37. The anode is

A) Negative electrode

B) Positive electrode

C) Neutral

D) Insulator

Answer: B

---

38. The cathode is

A) Positive

B) Negative

C) Neutral

D) Insulator

Answer: B

---

39. The unit of heat is

A) Joule

B) Watt

C) Pascal

D) Volt

Answer: A

---

40. Endothermic reaction

A) Releases heat

B) Absorbs heat

C) No heat

D) Produces light

Answer: B

41. Exothermic reaction

A) Absorbs heat

B) Releases heat

C) No heat

D) Freezes

Answer: B

---

42. The atomic number of carbon is

A) 4

B) 6

C) 8

D) 12

Answer: B

---

43. The symbol of potassium is

A) P

B) K

C) Pt

D) Po

Answer: B

---

44. The symbol of sodium is

A) Na

B) So

C) Sn

D) S

Answer: A

---

45. The molecular formula of methane is

A) CH_2

B) CH_3

C) CH_4

D) C_2H_4

Answer: C

---

46. Organic chemistry deals with

A) Metals

B) Carbon compounds

C) Salts

D) Gases

Answer: B

---

47. The simplest organic compound is

A) Methane

B) Ethane

C) Propane

D) Butane

Answer: A

---

48. Functional group in alcohol is

A) -COOH

B) -OH

C) -CHO

D) $-NH_2$

Answer: B

---

49. Functional group in carboxylic acid is

A) -OH

B) -COOH

C) -CHO

D) $-NH_2$

Answer: B

---

50. Isomers have

A) Same formula, different structure

B) Different formula

C) Same structure

D) Same atoms

Answer: A

---

51. Rate of reaction depends on

A) Temperature

B) Concentration

C) Catalyst

D) All of these

Answer: D

---

52. Equilibrium state means

A) Reaction stops

B) Forward = backward reaction

C) Only forward reaction

D) Only backward reaction

Answer: B

---

53. Le Chatelier's principle deals with

A) Equilibrium

B) Energy

C) Heat

D) Pressure

Answer: A

---

54. The unit of entropy is

A) J/K

B) J

C) K

D) W

Answer: A

---

55. Electrochemical cell converts

A) Chemical to electrical

B) Electrical to chemical

C) Heat to chemical

D) Light to heat

Answer: A

---

56. Oxidizing agent

A) Gains electrons

B) Loses electrons

C) Neutral

D) No change

Answer: A

---

57. Reducing agent

A) Gains electrons

B) Loses electrons

C) Neutral

D) No change

Answer: B

---

58. SI unit of conductivity is

A) S/m

B) Ω

C) V

D) A

Answer: A

---

59. The element in liquid state at room temperature

A) Iron

B) Mercury

C) Copper

D) Zinc

Answer: B

---

60. The common oxidation state of oxygen is

A) -2

B) +2

C) +1

D) -1

Answer: A